

WEB APPLICATION & ROBOTIC PROCESS AUTOMATION (RPA)

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SCENARIO 1 : AUTOMATED SOP INGESTION SYSTEM FOR
DHL LOGISTICS

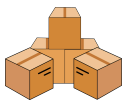
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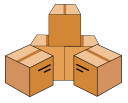


SECTION 1: CONTEXT & PROBLEM DESCRIPTION

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- Core Challenge: Regional DHL branches continuously submit Standard Operating Procedures (SOPs). Handling these manually causes data entry delays, typos, and operational bottlenecks.
- The Goal of Scenario 1: Automate file reading from local folders, enforce rigorous validation checking, and intercept duplicate submissions before they can pollute our database infrastructure.

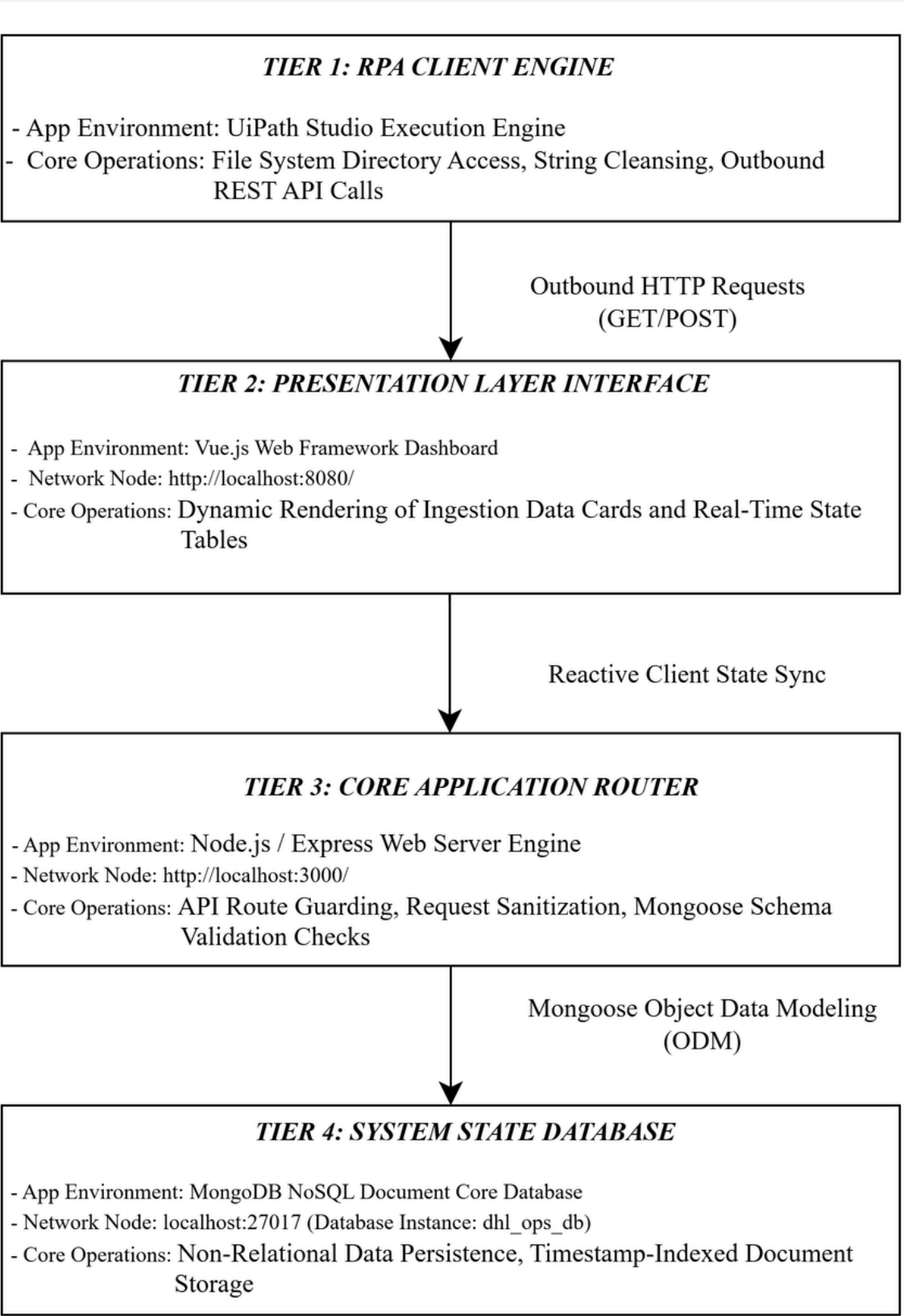


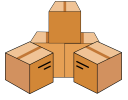


SECTION 2: THE 4-TIER SYSTEM ARCHITECTURE

This platform operates as a completely decoupled 4-Tier stack to separate concerns and guarantee system stability:

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SECTION 3: API & JSON PAYLOAD CONTRACTS

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The interface between the UiPath robot client and backend database is completely stateless and handles specific key-value parameters:

GET /api/articles (Master Log Retrieval):

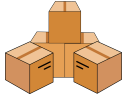
The UiPath robot sends a GET request here during startup to download a complete history list into its memory cache.

POST /api/articles (Document Ingestion):

If a document is proven unique, the robot triggers this request to push the new schema payload into the database.

Real Payload Validation Parameters:

- Instead of a generic text dump, our system processes specific business-relevant fields:
- sopId (The unique primary string identifier, e.g., SOP-003).
 - title (The core logistics descriptive string).
 - status (Enum values enforcing prioritization: 'Normal' or 'High').
 - lifecycleStatus (State tracking milestones: 'Draft', 'Reviewed', or 'Published').



SECTION 4: INSIDE THE UIPATH AUTOMATION LOGIC

The robot runs an optimized local process loop to save server processing costs:

01

Read & Clean
The robot loops through files in the target directory, isolates the file title, and executes `ExtractedContent.Trim()` in memory to strip out accidental whitespace padding.

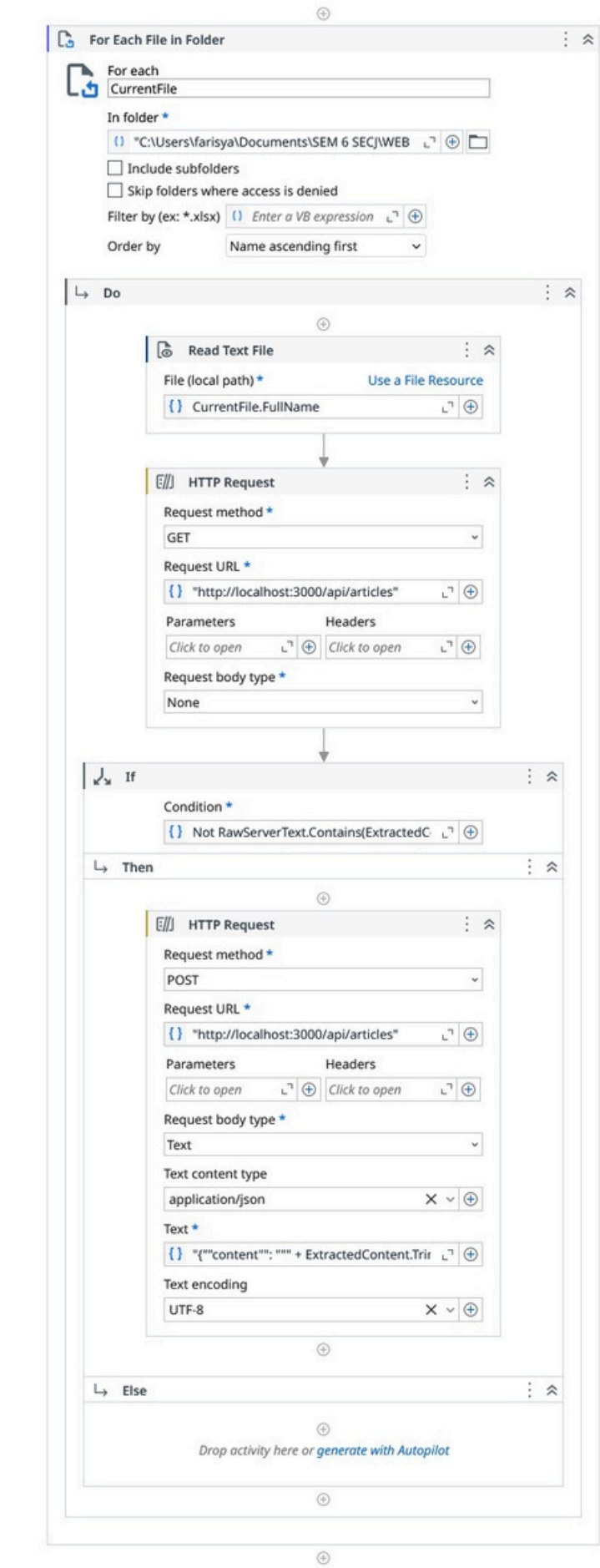
02

Fetch History
The robot calls our Express GET endpoint and saves the entire database text history array into a local variable named `RawServerText`.

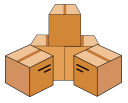
03

The Duplicate Gate
An If activity executes a strict string containment test: `Not RawServerText.Contains(CurrentFile_Title.Trim())`

- If True (Unique): It moves to Step 4 to initiate upload.
- If False (Duplicate): It immediately aborts the upload, prints a duplicate warning log to the console, and skips to the next file.
- Step 4: Sanitize & Upload: It replaces structural quote characters using `.Replace("''", "\\''")` to avoid JSON structural crashes, then fires a POST request to commit the structured record instantly.



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SECTION 5: SYSTEM VERIFICATION & TECHNICAL VALIDATION

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01

End-to-End Pipeline Validation

Empirical evaluation confirmed seamless integration across the entire distributed runtime network stack.

02

Schema Rigidity

Proved Mongoose middleware successfully intercepted and rejected any structurally invalid or malformed data payloads at the perimeter.

03

Server Execution Logging

Confirmed authenticated database connections to the dh1_cos_db instance and verified real-time audit generation ([AUDIT LOG SAVED] Action: CREATE_SOP).

04

State Synchronization

Reactive Vue.js frontend dynamically rendered automated outputs (SOP-001, SOP-002, SOP-003), preserving designated priority tiers and lifecycle tracking states.

05

Quantitative System Impact

Reactive Vue.js frontend dynamically rendered automated outputs (SOP-001, SOP-002, SOP-003), preserving designated priority tiers and lifecycle tracking states.

Schema Rigidity

```
14 // =====
15 // 1. DATABASE SCHEMA DEFINITION (Mongoose)
16 // =====
17 const ArticleSchema = new mongoose.Schema({
18   sopId: { type: String, required: true, unique: true },
19   title: { type: String, required: true },
20   creator: { type: String, default: 'System_Admin' },
21   status: { type: String, enum: ['Normal', 'High'], default: 'Normal' },
22   source: { type: String, default: 'MANUAL' },
23   // Explicitly defined lifecycleStatus with allowed array states in the schema
24   lifecycleStatus: {
25     type: String,
26     enum: ['Draft', 'Reviewed', 'Published'],
27     default: 'Draft'
28   }
29 }, { timestamps: true });
30
31 const Article = mongoose.model('Article', ArticleSchema);
32
33 // System Audit Logs Schema
34 const LogSchema = new mongoose.Schema({
35   action: { type: String, required: true }, // e.g., 'CREATE_SOP', 'UPDATE_SOP', 'DELETE_SOP'
36   details: { type: String, required: true },
37   timestamp: { type: String, required: true }
38 });
39
40 const Log = mongoose.model('Log', LogSchema);
41
```

Figure 1 : Backend source control workspace establishing the core Mongoose data validation schema and collection parameters.

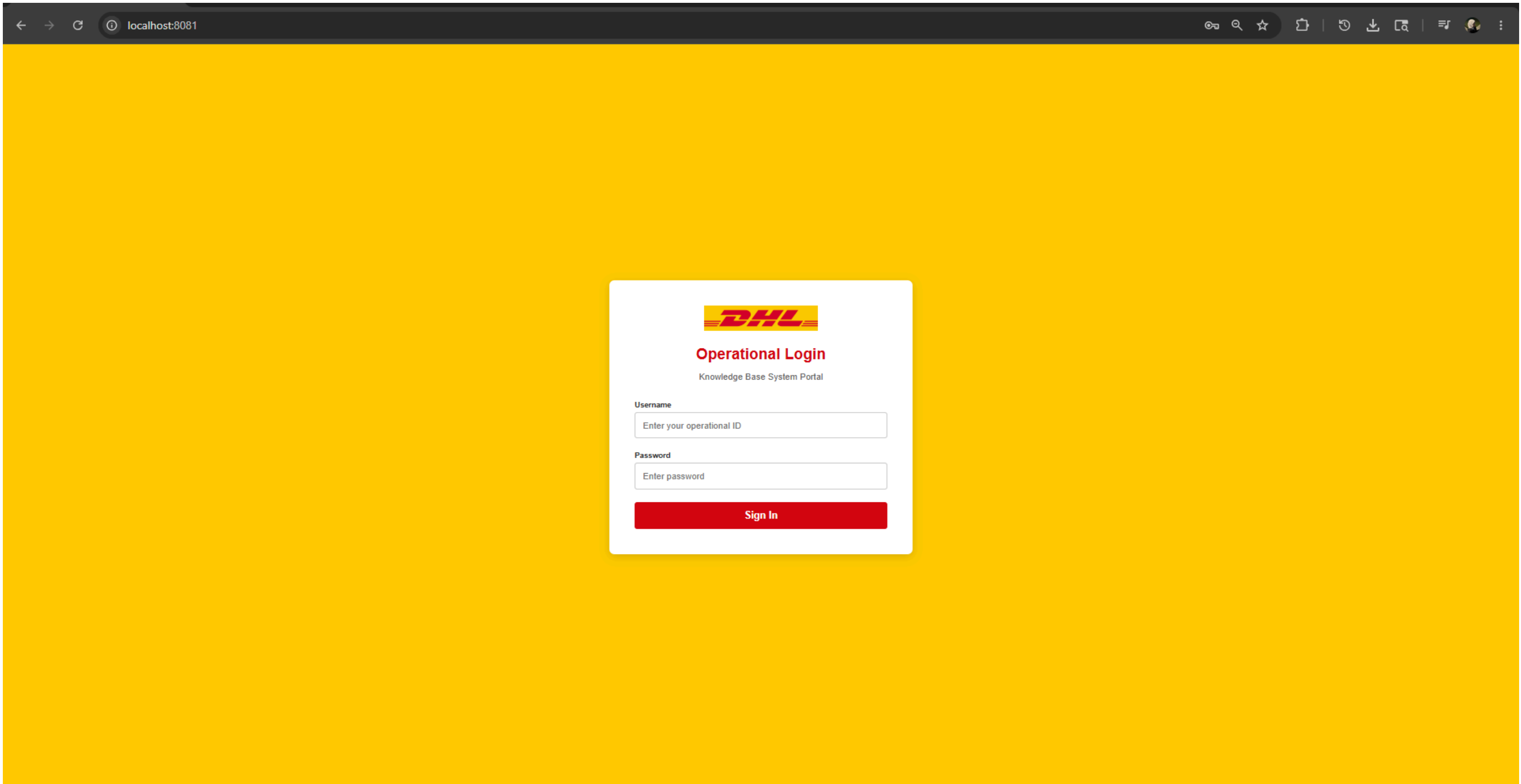
Server Execution Logging

```
=====
SUCCESS: Connected cleanly to MongoDB (dhl_ops_db)
Express routing infrastructure engine running on Port: 3000
=====
[AUDIT LOG SAVED] Action: CREATE_SOP | Details: Published new structural manifest node [SOP-003] - Title: "Urgent Cold-Chain Temperature Deviation Response Protocol for Pharmaceutical Air Freight"
(node:13824) [MONGOOSE] Warning: mongoose: the `new` option for `findOneAndUpdate()` and `findOneAndReplace()` is deprecated. Use `returnDocument: 'after'` instead.
(Use `node --trace-warnings ...` to show where the warning was created)
```

Figure 2: Live development console stream displaying successful application-tier initialization and active runtime handshake verification with the dhl_cos_db cluster instance.

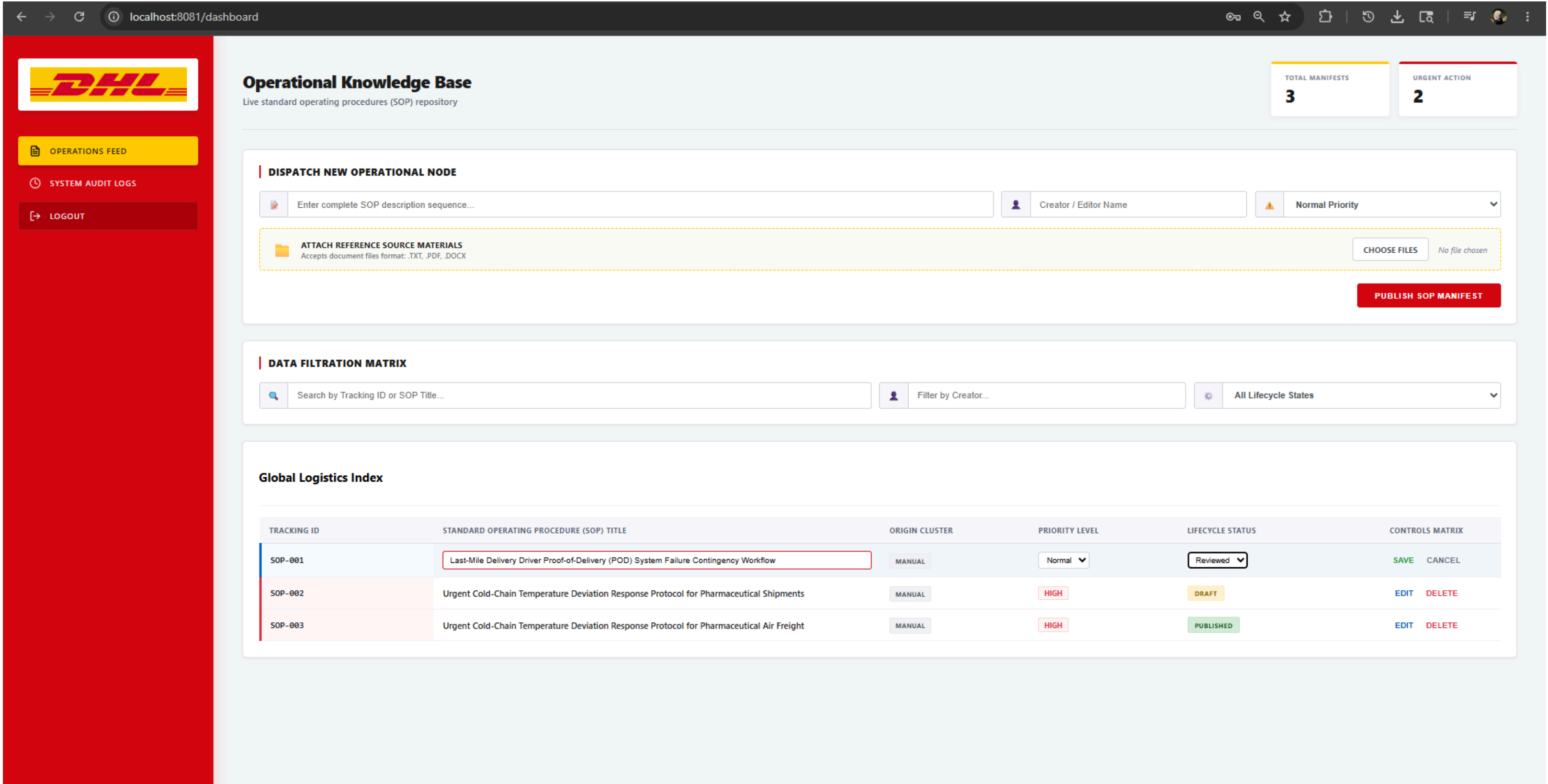
User Authentication Interface (Login Page)

Localized administrator authentication interface providing secure credential verification and session handshakes prior to dashboard application access.



The Active Production Management Dashboard

Figure 3 : Reactive web portal displaying the stable, persistent data states synchronized across the production cluster following the execution of automated test cases TC-01 through TC-03.



The System Audit Logs Table

Dedicated System Audit Logs panel recording operational transaction tracking histories and robotic workflow events.



OPERATIONS FEED

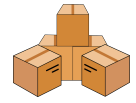
SYSTEM AUDIT LOGS

LOGOUT

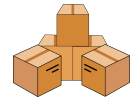
Database Audit Logging Matrix

System events ledger capturing CRUD transactions

TIMESTAMP	OPERATION	TRANSACTION DETAILS SEQUENCE
05/21/2026, 16:38:04	UPDATE_SOP	Modified record elements for [SOP-003]. Lifecycle status transition saved as: "Published"
05/21/2026, 16:38:00	UPDATE_SOP	Modified record elements for [SOP-003]. Lifecycle status transition saved as: "Reviewed"
05/21/2026, 16:37:45	CREATE_SOP	Published new structural manifest node [SOP-003] - Title: "Urgent Cold-Chain Temperature Deviation Response Protocol for Pharmaceutical Air Freight"
05/18/2026, 17:11:03	UPDATE_SOP	Modified record elements for [SOP-001]. Lifecycle status transition saved as: "Reviewed"
05/18/2026, 14:09:54	UPDATE_SOP	Modified record elements for [SOP-002]. Lifecycle status transition saved as: "Draft"
05/18/2026, 14:09:51	UPDATE_SOP	Modified record elements for [SOP-002]. Lifecycle status transition saved as: "Reviewed"
05/18/2026, 14:09:39	UPDATE_SOP	Modified record elements for [SOP-001]. Lifecycle status transition saved as: "Published"
05/18/2026, 14:09:35	UPDATE_SOP	Modified record elements for [SOP-001]. Lifecycle status transition saved as: "Published"
05/18/2026, 14:09:31	UPDATE_SOP	Modified record elements for [SOP-001]. Lifecycle status transition saved as: "Published"
05/18/2026, 14:09:27	UPDATE_SOP	Modified record elements for [SOP-001]. Lifecycle status transition saved as: "Reviewed"
05/18/2026, 14:09:14	CREATE_SOP	Published new structural manifest node [SOP-002] - Title: "Urgent Cold-Chain Temperature Deviation Response Protocol for Pharmaceutical Shipments"
05/18/2026, 14:07:37	UPDATE_SOP	Modified record elements for [SOP-001]. Lifecycle status transition saved as: "Draft"



DEMO



THANK YOU

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